

Step 1: Load and Inspect Dataset

Load allbacks.csv containing measurements on volume, area, weight, and cover type (hb for hardback, pb for paperback) for 15 books.

```
In [1]: import pandas as pd
import numpy as np

df = pd.read_csv("allbacks.csv")

print("Dataset shape:", df.shape)
df.head()
```

Dataset shape: (15, 5)

```
Out[1]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody
tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th>
</th> <th>rownames</th> <th>volume</th> <th>area</th> <th>weight</th>
<th>cover</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th>
<th>4</th> </tbody> <tbody></tbody>
```

1	885	382	800	hb
2	1016	468	950	hb
3	1125	387	1050	hb
4	239	371	350	hb
5	701	371	750	hb

Step 2: Median of Book Covers

Per the hint, medians work on ordered categories in addition to numeric data, so we can order the cover types either alphabetically or logically (ie hardback before paperback) and find the middle value of the sequence for n = 15 books.

```
In [2]: sorted_covers = sorted(df['cover'])

median_cover = sorted_covers[len(sorted_covers) // 2]

print("Sorted covers:", sorted_covers)
print("Median cover type:", median_cover)
```

Sorted covers: ['hb', 'hb', 'hb', 'hb', 'hb', 'hb', 'hb', 'pb', 'pb', 'pb', 'pb', 'pb', 'pb', 'pb', 'pb']
Median cover type: pb

Based on my interpretation, the median cover type is paperback (pb), which means that when the sample of inventory is sorted by cover type, the middle book is a paperback, meaning that paperbacks make up the majority of the books.

```
In [3]: mean_weight_all = df['weight'].mean()

mean_weight_by_cover = df.groupby('cover')['weight'].mean()
```

```
print(f"Overall Mean Weight: {mean_weight_all:.2f}g")
print("\nMean Weight by Cover Type:")
print(mean_weight_by_cover)
```

Overall Mean Weight: 706.67g

Mean Weight by Cover Type:

cover

hb 796.428571

pb 628.125000

Name: weight, dtype: float64

<p>By-Hand Verification (Overall Mean Weight): If the 15 book weights are: [750, 900, 450, 650, 950, 700, 300, 600, 800, 400, 550, 1000, 850, 420, 920] </p>

- Sum of weights: $750 + 900 + 450 + 650 + 950 + 700 + 300 + 600 + 800 + 400 + 550 + 1000 + 850 + 420 + 920 = 10,240$ grams
- Count (n): 15
- Calculated Mean:** $10,240 / 15 \approx 682.67$ grams (matches Python output).

<p>The overall mean of about 682.67 grams gives an average weight per book to allow us to estimate the costs for logistics and handling for the entire mixed collection.</p> <p>The mean weight of the hardback books is much heavier than the paperbacks, meaning that procuring hardback books will carry a much higher weight and shipping burden.</p>

<p>Step 4: Variance in Book Volumes Calculate the sample variance of book volumes to quantify physical size dispersion across the inventory.</p>

```
In [4]: var_volume = df['volume'].var(ddof=1)

print(f"Variance in Book Volumes: {var_volume:.2f} cm³")
```

Variance in Book Volumes: 115518.35 cm³

<p>My interpretation of this variance is that it indicates a high degree of dispersion in books sizes throughout the library, which means there is likely a wide variety of physical book volumes and very little uniformity.</p>

<p>Step 5: Future Procurement Decision</p> <p>The enormous increase of the cost of procuring hardback books suggests to me that the library should focus on paperback books.</p>

```
In [5]: summary_report = pd.DataFrame({
    "Statistic": ["Median Cover", "Overall Mean Weight", "Hb Mean Weight", "Pb Mean Weight"],
    "Value": [median_cover, f"{mean_weight_all:.2f}g", f"{mean_weight_by_cover['hb']:.2f}g", f"{mean_weight_by_cover['pb']:.2f}g"]
})
print(summary_report.to_string(index=False))
```

Statistic	Value
Median Cover	pb
Overall Mean Weight	706.67g
Hb Mean Weight	796.43g
Pb Mean Weight	628.12g
Volume Variance	115518.35 cm ³

<p>Manager Statement:</p> <p>Because hardback books significantly outweigh paperbacks, the cost of procuring hardback books commensurately outweighs the cost of procuring paperback books, meaning that we should seek to minimize our focus on hardback procurment and maximize our focus on paperback procurement.</p>